PIAGGIO WOULD LIKE TO THANK YOU

for choosing one of its products. We have prepared this manual to help you to get the very best from your vehicle. Please read it carefully before riding the vehicle for the first time. It contains information, tips and precautions for using your vehicle. It also describes features, details and devices to assure you that you have made the right choice. We believe that if you follow our suggestions, you will soon get to know your new scooter and it will serve you well for a long time to come. This booklet forms an integral part of the vehicle; should the vehicle be sold, it must be transferred to the new owner.

ZIP 50 - 100 4T

PIAGGIO®

Ed. 03_10/2012

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The instructions given in this booklet are intended to provide a clear, simple guide to using your scooter; details are also given of routine maintenance procedures and regular checks that should be carried out on the vehicle at and **Authorized PIAGGIO Dealer or Service Centre**. The booklet also contains instructions for simple repairs. Any operations not specifically described in this booklet require the use of special tools and/or particular technical knowledge: to carry out these operations refer to any **authorized PIAGGIO Dealer of Service Centre**.

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Personal safety

Failure to completely observe these instructions will result in serious risk of personal injury.

Safeguarding the environment

Sections marked with this symbol indicate the correct use of the vehicle to prevent damaging the environment.

Vehicle intactness

The incomplete or non-observance of these regulations leads to the risk of serious damage to the vehicle and sometimes even the invalidity of the guarantee.

The signs that you see on this page are very important. They are used to highlight parts of the booklet that should be read with particular care. The different symbols are used to make each topic in the manual simple and quick to locate.

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INDEX

VEHICLE 7
Dashboard 8
Keyswitch 8
Lockingthesteeringwheel 9
Switchdirectionindicators 9
Hornbutton 10
Lightswitch 10
Start-upbutton 11
Openingthesaddle 11
Keys 13
Identification 14
USE 15
Checks 16
Refuelling 16
Tyrepressure 17
Runningin 18
Startinguptheengine 19
Difficultstartup 20
Stoppingtheengine 21
Automatictransmission 21
Safedriving 22
MAINTENANCE 25
Engineoillevel 26
Engineoillevelcheck 26
Engineoiltop-up 27
Engineoilchange 27
Huboillevel 28
Tyres 29
Sparkplugdismantlement 30
Removingtheairfilter 31
Secondaryairsystem 32
Checkingthebrakeoillevel 33
Battery 34
Long periods of inactivity 35
Fuses 36
Frontlightgroup 37
Headlightadjustment 38
Frontdirectionindicators 38
Rearopticalunit 39
Rearturnindicators 40
Rear-viewmirrors 41
Idleadjustment 41
Frontandreardiscbrake 42
Reardrumbrake 43
Puncture 44
delveicolo 45
Cleaningthevehicle 45
TECHNICALDATA 51
Toolkit 58
SPAREPARTSANDACCESSORIES 59
Warnings 60
SCHEDULEDMAINTENANCE 63
Scheduledservicingtable 64

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PIAGGIO

Chap. 01
Vehicle

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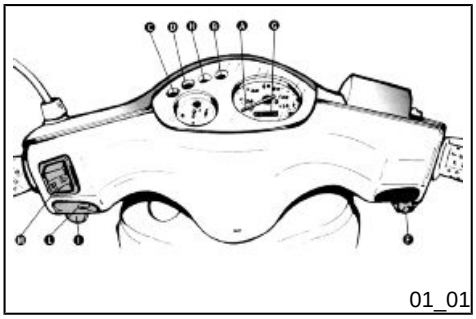


01_01

Dashboard (01_01)

A = Speedometer

B = Low fuel warning light

C = Turn indicator warning light

D = Preset warning light

F = Starter button

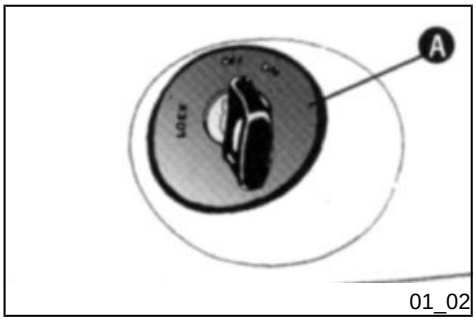
G = Odometer

H = High-beam headlight warning light

I = Horn button

L = Turn indicator switch

M = Headlight switch



01_02

Keyswitch (01_02)

LOCK = Ignition to earth, extractable key, steering lock engaged.

OFF = Ignition to earth, extractable key, steering lock disengaged.

ON = Ready to start, steering lock disengaged, key cannot be extracted.

In the position«**ON**» and «**OFF**» it is possible to open the front glove-box pushing the key switch, in position **LOCK** the glove-box is closed.

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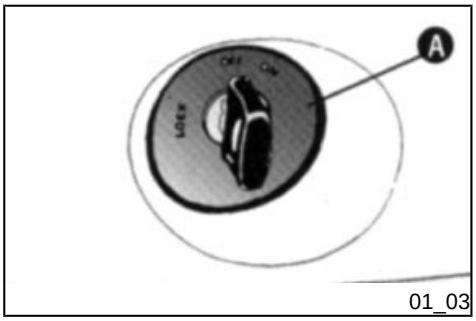


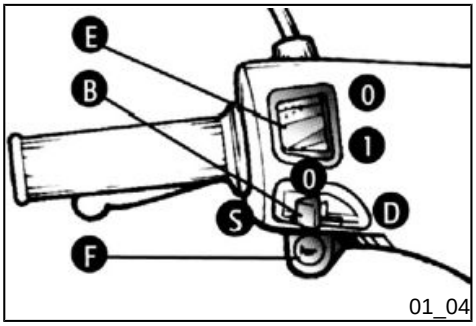
01_03

Lockingthesteeringwheel (01_03)

Turn the handlebar to the left (as far as it will go), turn the key to «**LOCK**» and remove the key.

CAUTION

DO NOT TURN THE KEY TO «LOCK» OR «KEY OFF» WHILE RIDING.



01_04

Switchdirectionindicators (01_04)

0 = Turn indicators off.

D = Turn indicators on (right side)

S = Turn indicators on (left side)

The positions «**D**» and «**S**» are unstable. The switch lever «**B**» always returns to «**0**». Push the lever to switch off the turn indicators.

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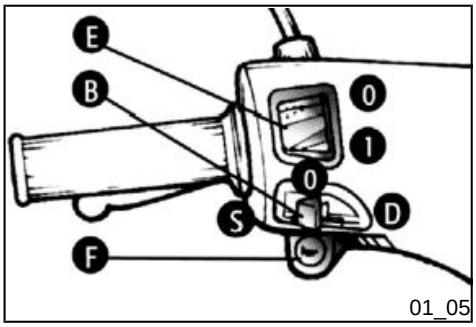
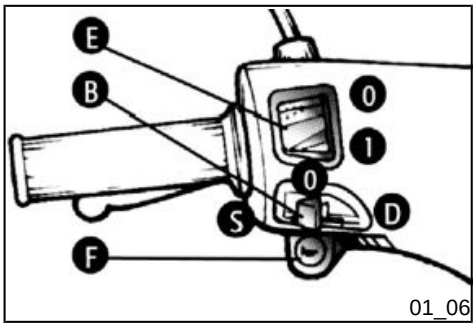


01_05

Hornbutton (01_05)

Horn button «**F**»



01_06

Lightswitch (01_06)

0 = Low-beam and tail light

1 = High-beam and tail light

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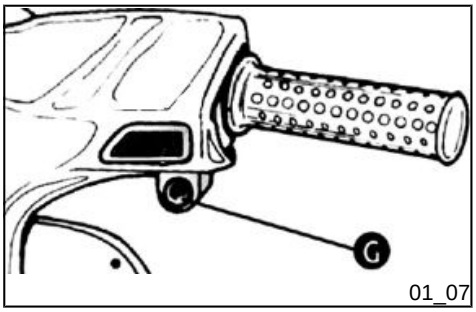

01_07

Start-upbutton (01_07)

Starter button "**G**"

Openingthesaddle (01_08, 01_09, 01_10)

Insert the key into the saddle lock «**A**», turn it counterclockwise and tip the saddle forward.

In this scooter, the saddle height can be adjusted to three positions thanks to a rack «**C**» assembled on the saddle hinge and to the stroke buffers «**B**».

Adjustments:

rack low position-buffer position **1**

rack medium height-buffer position **2**

rack high position- buffer position **3**

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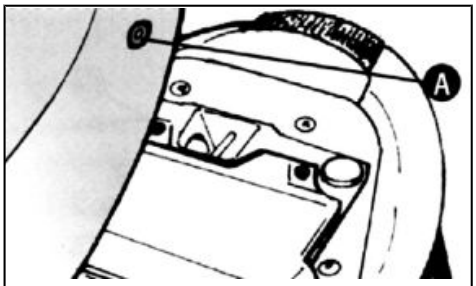
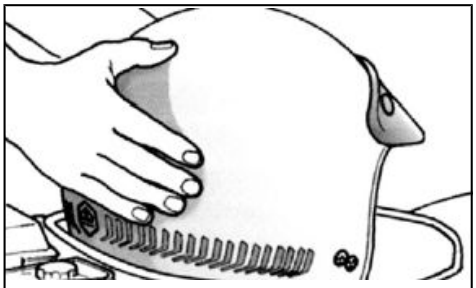


01_08


01_09

Insert the key in lock «**A**», turn it anticlockwise and tip the saddle forward.
After having tipped the saddle, place the helmet as shown in the figure.

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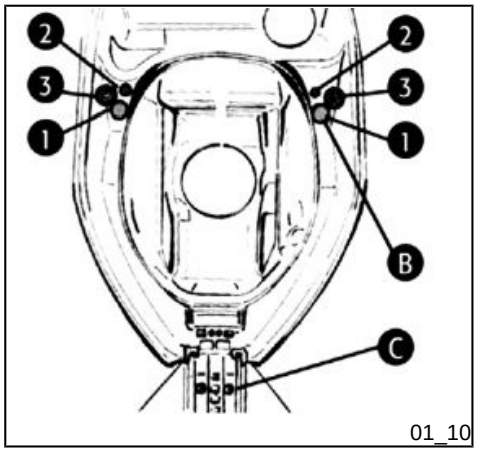
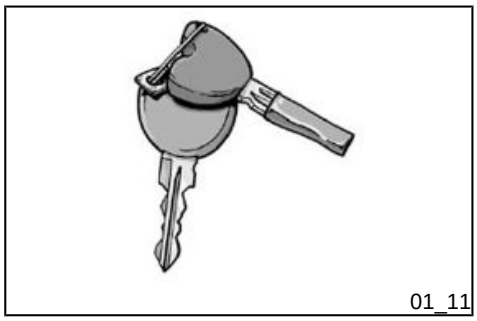


01_10


01_11

Keys (01_11)

The vehicle is supplied with two keys (one spare) which serve to start the engine and unlock the saddle compartment. The keys are accompanied by a tag marked with the identification code to be quoted when ordering duplicates.

WARNING

WE RECOMMEND KEEPING THE DUPLICATE KEY TOGETHER WITH ITS CODE IN A SAFE PLACE AND NOT ON THE VEHICLE

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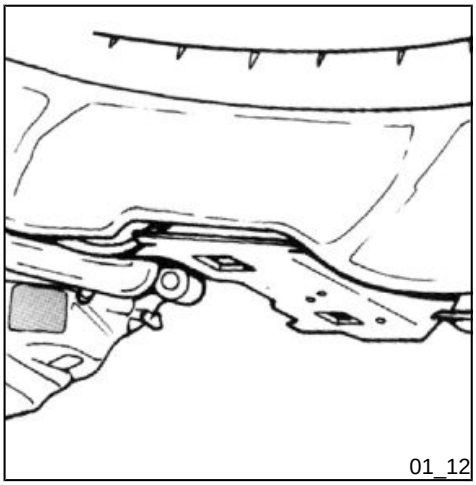
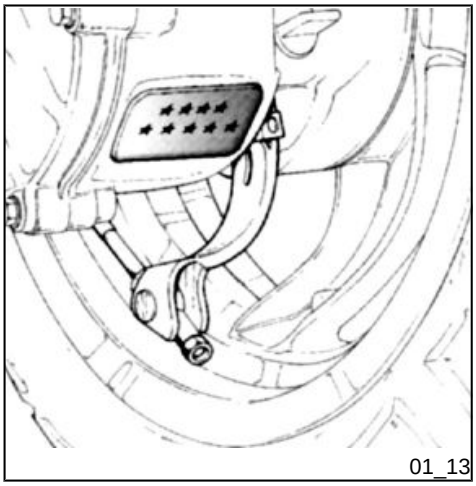


01_12



01_13

Identification (01_12, 01_13)

The identification numbers consist of a prefix stamped on the chassis and on the engine, followed by a number. They must be quoted when ordering spare parts. We recommend that you check that the prefix and chassis number stamped on the vehicle correspond with those in the vehicle documents.

CAUTION

PLEASE REMIND THAT ALTERING IDENTIFICATION REGISTRATION NUMBERS CAN LEAD TO SERIOUS PENAL SANCTIONS (IMPOUNDING OF THE VEHICLE, ETC.).

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ZIP 50 - 100 4T

PIAGGIO

Chap. 02
Use

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Checks

Before using the vehicle, check:

1. that the fuel tank is full.

2. The oil level in the rear hub.

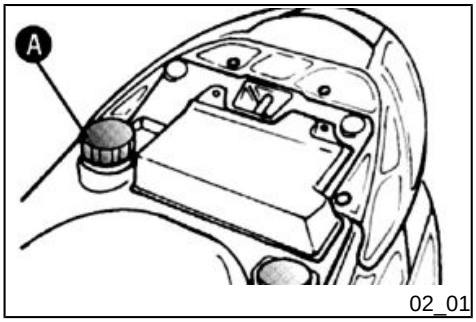
3. engine oil level (see section «Engine oil level»).

4. That tyres are properly inflated.

5. correct functioning of headlights, rear taillight and turn indicators.

6. The correct functioning of the front and rear brakes.

7. The fluid level in the front brake pump tank.



02_01

Refuelling (02_01, 02_02)

Fill fuel tank«**A**» with unleaded petrol (minimum octane rating = 95).
When the fuel reaches the reserve level, the warning light fitted on the instrument panel lights up.

CAUTION

SHUT OFF THE ENGINE BEFORE REFUELLING WITH PETROL. PETROL IS HIGHLY FLAMMABLE. DO NOT LET PETROL SPILL FROM THE TANK OR WHILE REFUELLING

CAUTION

DO NOT BRING NAKED FLAMES OR CIGARETTES NEAR THE MOUTH OF THE FUEL TANK: FIRE HAZARD. ALSO AVOID INHALING HARMFUL VAPOURS.

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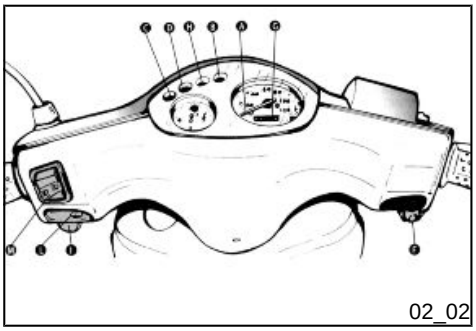

02_02

CAUTION

USING OILS OTHER THAN THOSE RECOMMENDED CAN SHORTEN THE LIFE OF THE ENGINE.

Characteristic

Fuel tank

In plastic, with a capacity of ~ 7.5 l (approximate value, including ~ 1.2 l reserve)

Main and supplementary fuel tank (including ~ 1.2 l reserve) (100 cc)

~ 7 l

Tyrepressure

CAUTION

TYRE PRESSURE SHOULD BE CHECKED WHEN TYRES ARE COLD.INCORRECT TYRE PRESSURE CAUSES ABNORMAL TYRE WEAR AND MAKES RIDING DANGEROUS.

TYRES MUST BE REPLACED WHEN THE TREAD REACHES THE WEAR LIMITS SET FORTH BY LAW.

Characteristic

Front tyre pressure

Front wheel: 1.3 bar

Rear tyre pressure

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Rear wheel: 1.6 bar

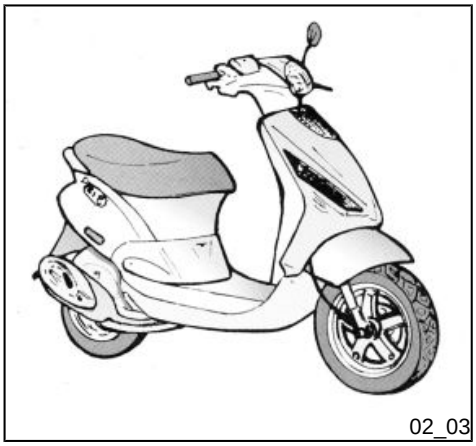
Rear tyre pressure (rider and luggage)

Rear wheel (rider and luggage): 1.8 bar

Tyre pressure (100 cc)

Front wheel: 1.3 bar

Rear wheel: 1.6 ÷ 1.8 bar



02_03

Runningin (02_03)

WARNING

DURING THE FIRST 1000 KM DO NOT RIDE THE VEHICLE OVER 80% OF ITS MAXIMUM SPEED. AVOID TWISTING THE THROTTLE GRIP FULLY OR KEEPING A CONSTANT SPEED ALONG LONG SECTIONS OF ROAD. AFTER THE FIRST 1000 KM, GRADUALLY INCREASE SPEED UNTIL REACHING THE MAXIMUM PERFORMANCE.

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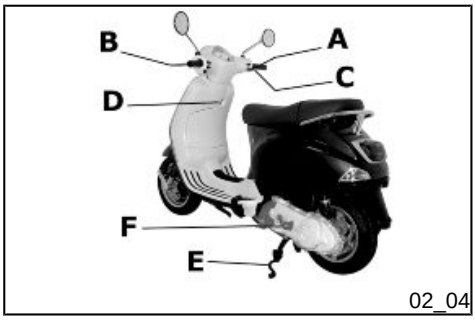

02_04

Startinguptheengine (02_04)

The vehicle is fitted with direct drive automatic transmission. Therefore, always start the engine with the throttle grip slightly twisted; to start-off from stationary position, progressively twist the throttle grip.

The vehicle is equipped with a fuel valve and a starter which switch on automatically as soon as the engine is started.

To start it up, it is necessary to:

1: Put the scooter on its stand "**E**"; check that the rear wheel is off the ground.

2: Keep the throttle grip slightly twisted.

3: Insert the key in ignition switch «**D**» and turn it to «**ON**»; when turning the key to «**ON**», the fuel reserve warning light turns on for a few seconds indicating the correct functioning of the bulb.

4: Push the starter button «**A**» after pulling the rear brake lever «**B»** or the front brake lever «**C**».

CAUTION

DO NOT CARRY OUT THESE OPERATIONS IN CLOSED AREAS SINCE EXHAUST GASES ARE TOXIC.

CAUTION

ALWAYS PLACE THE VEHICLE ON ITS STAND BEFORE KICK STARTING.

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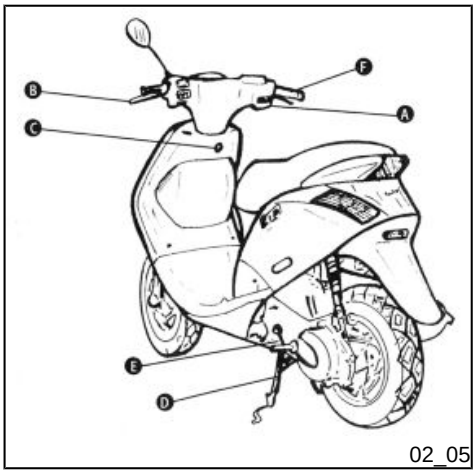

02_05

Difficultstartup (02_05)

If there is a problem you can follow the instructions below:

1. Engine flooded. Place the vehicle on its centre stand and check that the rear wheel is off the ground. Open the throttle fully and press the starter button for five seconds and then stop for five seconds. If the engine does not start after a few attempts, let the engine sit for a few minutes and then repeat the above operations. In any case do not operate the starter motor longer than 20" in an attempt to start the engine.

2. In the case of battery inefficiency. Place the vehicle on stand «**E**»; make sure that the rear wheel is off the ground, turn the key switch «**D**» to «**ON**».

3. Empty fuel tank. After refuelling the scooter, start the engine by pressing the starter button «**A**» with the throttle at a minimum to provide maximum aspiration for the tap. If the scooter fails to start even after taking the steps described above, contact an **Authorised Piaggio Service Centre.**

CAUTION

ALWAYS PLACE THE VEHICLE ON ITS STAND BEFORE KICK STARTING.

WARNING

TAMPERING MAY CAUSE SERIOUS ENGINE MALFUNCTION.

AS THE CARBURETTOR IS FITTED WITH AN ACCELERATOR PUMP DO NOT ENGAGE THE ACCELERATOR WITH THE ENGINE OFF IN ORDER TO AVOID FLOODING SPECIALLY IN HOT WEATHER.

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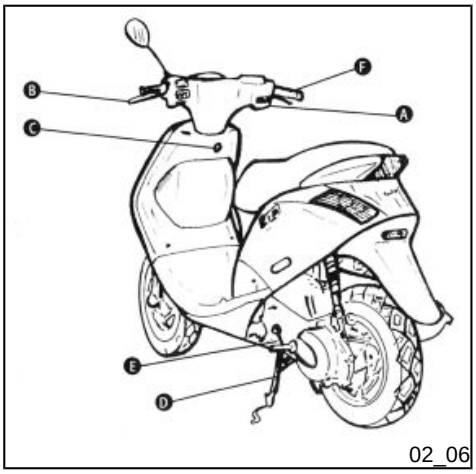

02_06

Stoppingtheengine (02_06)

Stop acceleration, then turn the key switch «**D**» to «**OFF** » to stop the engine (extractable key).

02_07

Automatictransmission (02_07)

To ensure simple, pleasurable riding, the vehicle is equipped with automatic transmission with regulator and centrifugal clutch. The system is designed to give the best possible performance in terms of both acceleration and consumption, on level ground and uphill, thanks to the adjustments made to engine speed and transmitted torque. If you have to stop on an uphill slope (traffic lights, traffic jam, etc.) only use the brake to keep the vehicle still, leaving the motor running at idling speed. Using the motor to keep the vehicle still can cause the clutch to overheat. This problem is due to the friction of the clutch parts on the clutch bell. It is therefore recommended to avoid conditions of prolonged clutch slippage leading to clutch overheating (for example, as well as the situation described above, riding uphill fully laden on steep slopes or starting off on slopes greater than 25%, etc.):

1. Do not continue riding in such conditions.

2. Let the clutch cool down with the motor at idling speed for a few minutes

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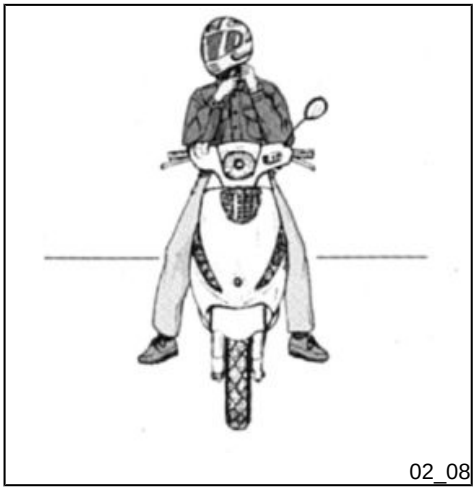
Safedriving (02_08)

WARNING

SOME SIMPLE TIPS ARE PROVIDED BELOW WHICH WILL ENABLE YOU TO USE YOUR VEHICLE ON A DAILY BASIS MORE EASILY AND SAFELY.

<



02_08

Your ability and your knowledge of the vehicle form the basis of safe riding. We recommend trying out the vehicle in traffic-free zones to get to know your vehicle completely.

ALWAYS DRIVE WITHIN YOUR LIMITS

1. Before riding off, remember to put on your helmet and fasten it correctly.

2. Reduce speed and ride cautiously on uneven roads.

3. Remember that after riding on a long stretch of wet road without using the brakes, the braking effect is initially lower. Under these conditions, it is a good idea to operate the brakes from time to time.

4. Do not brake hard on a wet surface, on dirt tracks or on any slippery road surface.

5. If you have to brake, use both brakes in order to divide the braking action between both wheels.

6. Avoid starting off by mounting the vehicle while it is still resting on its stand. In any case, the rear wheel should not be turning when it comes into contact with the ground, in order to avoid abrupt departures.

7. If the vehicle is used on roads covered with sand, mud, snow mixed with salt, etc., clean the brake disc frequently with mild detergent in order to prevent abrasive substances from building up within the holes, which can result in early wear of the brake pads.

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8. Any elaboration that modifies the vehicle's performances, such as tampering with original structural parts is strictly forbidden by law, and renders the vehicle not conforming to the approved type and therefor dangerous to ride.

CAUTION

RIDING UNDER THE INFLUENCE OF ALCOHOL, DRUGS OR CERTAIN MEDICINES CAN BE EXTREMELY DANGEROUS FOR ONESELF AND FOR OTHERS.

CAUTION

ANY ELABORATION THAT MODIFIES THE VEHICLE'S PERFORMANCES, SUCH AS TAMPERING WITH ORIGINAL STRUCTURAL PARTS IS STRICTLY FORBIDDEN BY LAW, AND RENDERS THE VEHICLE NO LONGER CONFORMING TO THE APPROVED TYPE AND DANGEROUS FOR RIDING.

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ZIP 50 - 100 4T

PIAGGIO

Chap. 03
Maintenance

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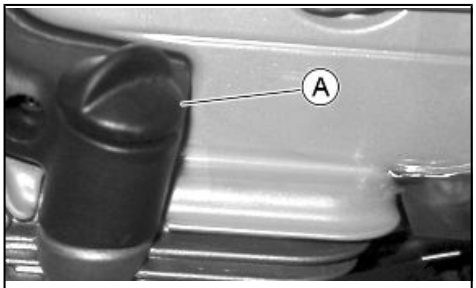
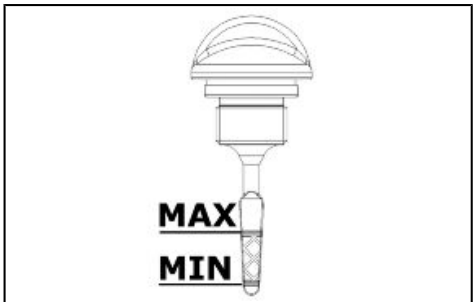
Engineoillevel

In 4T engines, the engine oil is used to lubricate the distribution elements, the bench bearings and the thermal group. **An insufficient quantity of oil can seriously damage the engine.** In all 4T engines the decay of oil characteristics as well as some consumption is normal. Consumption can particularly reflect the conditions of use (i.e. when driving at 'full acceleration' all the time, oil consumption increases).
In order to avoid problems, it is advisable to control oil level every time the vehicle is used.



03_01



03_02

Engineoillevelcheck (03_01, 03_02)

Check oil level every time the vehicle is used, with the engine cold. The level should be between the MAX and MIN marks on the oil level dipstick «**A** » shown in the figure; the stand should be firm on an even surface during the check.

If the check is carried out after the vehicle has been used, and therefore with a hot engine, the level line will be lower; in order to carry out a correct check, wait at least 10 minutes after the engine has been stopped so as to get the correct level.

Check by pushing the dipstick all the way into the hole.

CAUTION

RUNNING THE ENGINE WITH INSUFFICIENT LUBRICATION OR WITH THE INADEQUATE LUBRICANTS ACCELERATES WEAR AND TEAR OF THE MOVING PARTS AND CAN CAUSE IRRETRIEVABLE DAMAGE.

Characteristic

Engine oil

Capacity: ~ 850 cm³

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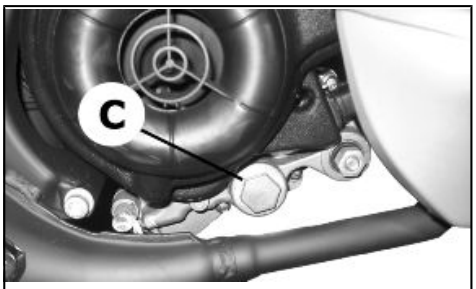
Engineoiltop-up

Always check oil level before carrying out top-ups and add oil **without exceeding the MAX level**. An engine oil check-up and top-up should be carried out every 3,000 km at an **Authorised Piaggio Service Centre**.



03_03

Engineoilchange (03_03)

Change oil and clean the filter after every 6,000 km at an **Authorised Service Centre**. Empty the engine by draining the oil through drainage cap «**C**» with a cold engine. First unscrew the cap-dipstick used to check engine oil level, then cap «**C**» and take out the mesh filter. After cleaning and mounting again the mesh filter, screw cap «**C**» and fill with oil the hole of the cap-dipstick used to check engine oil level. Then start the vehicle, leave it running for a few minutes and switch it off: 5 minutes later check oil level and, if necessary, add oil **without exceeding the MAX level**. Only use new oil of the recommend type for oil top-ups and change.

CAUTION

USED OILS CONTAIN SUBSTANCES HARMFUL TO THE ENVIRONMENT. FOR OIL CHANGE, CONTACT AN AUTHORISED SERVICE CENTRE WHICH IS EQUIPPED TO DISPOSE OF USED OILS IN AN ENVIRONMENTALLY FRIENDLY AND LEGAL WAY.

Recommended products

eni i-Ride PG 5W-40

Synthetic based lubricant for high-performance four-stroke engines.
JASO MA, MA2 - API SL - ACEA A3

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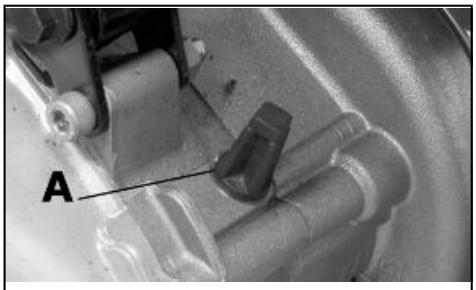
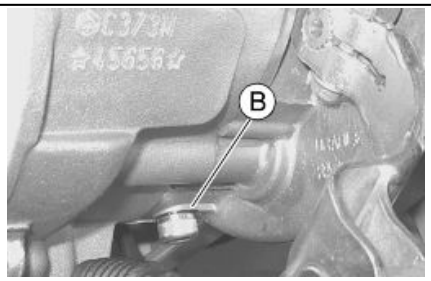
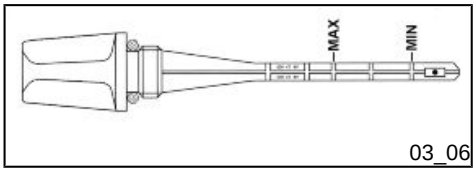


03_04



03_05



03_06

Huboillevel (03_04, 03_05, 03_06)

To check the hub oil level, proceed as follows:

1. Park the scooter on level ground and rest it on its stand.
2. Unscrew the oil dipstick «**A**», dry it with a clean cloth and then reinsert it, **screwing it fully into place**;
3. Unscrew the dipstick again and check that the oil level just reaches the **2nd notch from the bottom**;
4. Screw the dipstick back into place completely.

The screw «**B**» is the hub oil drainage plug.

CAUTION

RUNNING THE ENGINE WITH INSUFFICIENT LUBRICATION OR WITH THE INADEQUATE LUBRICANTS ACCELERATES WEAR AND TEAR OF THE MOVING PARTS AND CAN CAUSE IRRETRIEVABLE DAMAGE.

CAUTION

USED OILS CONTAIN SUBSTANCES HARMFUL TO THE ENVIRONMENT. FOR OIL REPLACEMENT, CONTACT AN AUTHORISED SERVICE CENTRE WHICH IS EQUIPPED TO DISPOSE OF USED OILS IN AN ENVIRONMENTALLY FRIENDLY AND LEGAL WAY.

N.B.

THE NOTCHES ON THE HUB OIL LEVEL DIPSTICK, EXCEPT THE ONE INDICATING THE MAXIMUM AND MINIMUM LEVEL, REFER TO OTHER MODELS BY THE MANUFACTURER AND HAVE NO SPECIFIC FUNCTION FOR THIS MODEL.

Recommended products

AGIP GEAR SAE 80W-90

Lubricant for gearboxes and transmissions.

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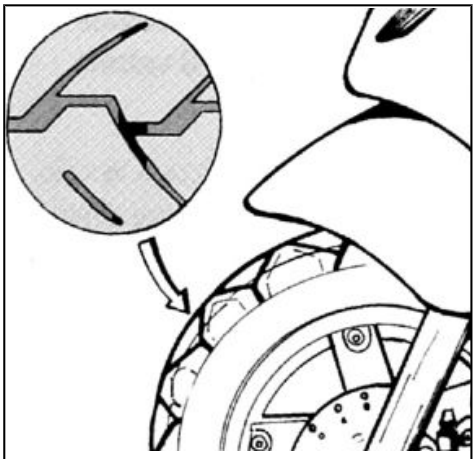
API GL-4

Characteristic

Rear hub oil

Quantity: approx. 85 cc



03_07

Tyres (03_07)

Periodically check the inflation pressure of each tyre. Tyres have tread wear indicators and must be replaced as soon as these indicators are visible on the tread; should this occur, go to an authorised workshop or to a workshop properly equipped for refitting.

CAUTION

TYRE PRESSURE SHOULD BE CHECKED WHEN TYRES ARE COLD.INCORRECT TYRE PRESSURE CAUSES ABNORMAL TYRE WEAR AND MAKES RIDING DANGEROUS.

TYRES MUST BE REPLACED WHEN THE TREAD REACHES THE WEAR LIMITS SET FORTH BY LAW.

Characteristic

Front tyre pressure

Front wheel: 1.3 bar

Tyre pressure (100 cc)

Front wheel: 1.3 bar

Rear wheel: 1.6 ÷ 1.8 bar

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Sparkplugdismantlement (03_08)

Remove the 3 screws «**A**», and the hatch too «**B**» sliding it upwards. Loosen the spark plug HV wire and remove the spark plugs with the box-spanners.

When refitting the plug, tighten it by hand, being sure to insert it at the right angle; Use the wrench only to fasten it. When refitting the hatch, follow the procedure in the reverse order, taking care that the teeth are introduced into the corresponding places on the upper cover.

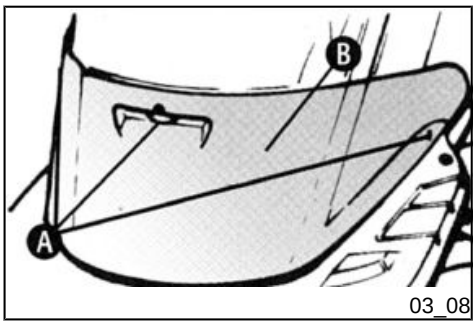
WARNING

THE SPARK PLUG MUST BE REMOVED WHEN THE ENGINE IS COLD. THE USE OF A SPARK PLUG WITH THERMAL GRADE OR THREAD OTHER THAN THE INDICATED TYPE (SEE «DATA» SECTION) CAN SERIOUSLY DAMAGE THE ENGINE. REPLACE SPARK PLUGS AT THE INTERVALS INDICATED IN THE SCHEDULED MAINTENANCE TABLE.

Characteristic

Spark plug

NGK CR8EB



03_08

Version 50 4T

Characteristic

Alternative spark plug

DENSO U24ESR-NB

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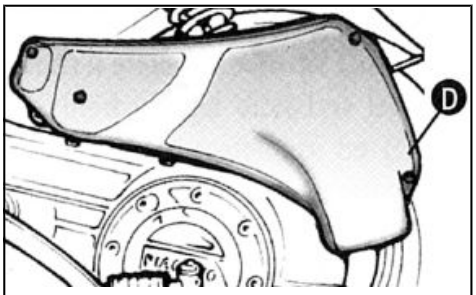

03_09

Removingtheairfilter (03_09)

Take out the air cleaner cover «**D**» after having unscrewed the 6 fixing screws, then take out the filter. Clean with water and shampoo, then dry with compressed air and immerse in a mix of oil and petrol at a ratio of 50%. Then gently squeeze the filter element between your hands, leave it to dry and mount it again.

CAUTION

IN CASE OF RIDING ON DUSTY ROADS IT IS ADVISABLE TO CLEAN THE AIR FILTER MORE FREQUENTLY THAN INDICATED IN THE RELEVANT CHAPTER ON SCHEDULED MAINTENANCE.

Recommended products

AGIP FILTER OIL

Special product for the treatment of foam filters.
-

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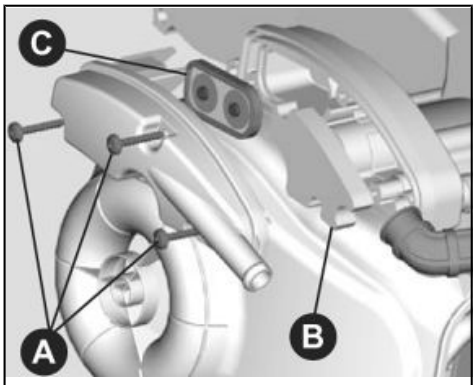

03_10

Secondaryairsystem (03_10)

Remove the three screws «**A**» on the cover of the secondary air system housing. Detach the housing cover and remove the filter and the sponge «**B**». wash the sponge with water and a neutral soap, dry with a clean cloth and short blasts of compressed air. While cleaning the filter, check the good condition of the reed valve **C** and mount it back in place on the housing.

Before closing the cover of the SAS housing, check the good condition of the sealing O-ring; replace it if damaged or deformed.

N.B.

The reed valve can be inserted in only one direction on the SAS housing

CAUTION

CONTACT AN AUTHORISED PIAGGIO SERVICE CENTRE TO CARRY OUT THESE OPERATIONS.

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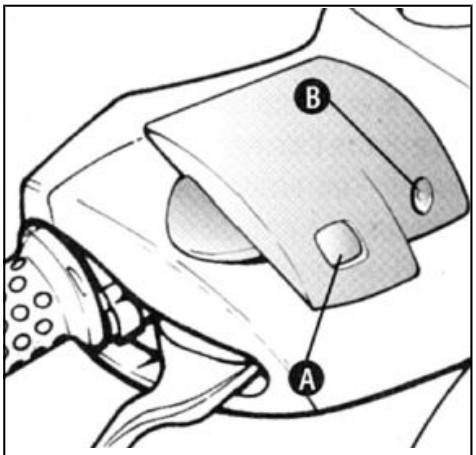
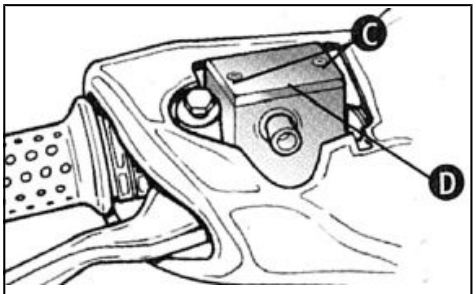


03_11



03_12

Checkingthebrakeoillevel (03_11, 03_12)

The brake fluid reservoir is fitted with a sight-glass "**A**"; the amount of brake fluid seen through this glass shows the level of fluid contained in the reservoir.

When the sight glass «**A**» is full, the level inside the reservoir exceeds the Min mark, when it is partially full, the level is slightly below the Min, when it is completely empty, the level is below the minimum.

The brake fluid level may fall due to wear on the brake pads. Should the level be below the minimum, the customer is advised to contact an **Authorised Piaggio Service Centre** for a thorough check of the braking system. To adjust the level follow the instructions below. Unscrew the screw «**B**» and remove the plastic cover, unscrew the 2 screws «**C**», lift the brake fluid reservoir cover«**D**» and add as much fluid as necessary, (the fluid level must always be above the minimum level mark).

Check these levels with the reservoir in a horizontal position, i.e. with the handlebars straight and the scooter perfectly level. Do not tilt the vehicle.

CAUTION

TOP-UPS SHOULD ONLY BE CARRIED OUT WITH DOT4 CLASSIFIED BRAKE FLUID.

CAUTION

THE BRAKING CIRCUIT FLUID IS HIGHLY CORROSIVE. THEREFORE, WHEN TOPPING UP, AVOID LETTING IT COME INTO CONTACT WITH THE PAINTED PARTS OF THE VEHICLE. THE BRAKING CIRCUIT FLUID IS HYGROSCOPIC, THAT IS, IT ABSORBS HUMIDITY FROM THE SURROUNDING AIR. IF MOISTURE CONTAINED IN THE BRAKE FLUID EXCEEDS A CERTAIN VALUE, THIS WILL RESULT IN INEFFICIENT BRAKING.

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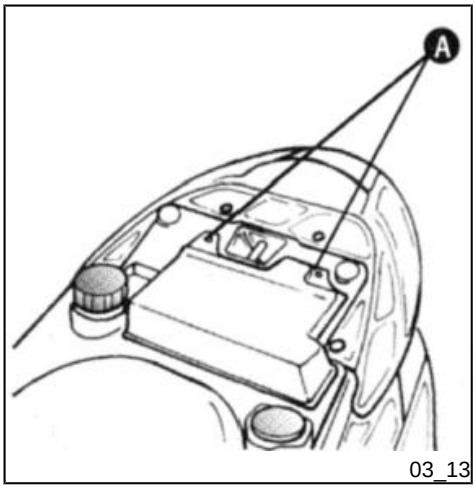
WARNING

IN NORMAL CLIMATIC CONDITIONS IT IS ADVISABLE TO REPLACE THE ABOVE-MENTIONED FLUID EVERY 2 YEAR. NEVER USE BRAKE FLUID CONTAINED IN CONTAINERS WHICH ARE ALREADY OPEN OR PARTIALLY USED.



03_13

Battery (03_13)

To have access to the battery it is necessary to remove the saddle («Keys» section) and the access hatch to the battery compartment unscrewing the star-head screws «**A**» shown in the figure

CAUTION

IN ORDER TO AVOID DAMAGING THE ELECTRICAL SYSTEM, NEVER DISCONNECT THE WIRING WHILE THE ENGINE IS RUNNING. DO NOT TIP THE VEHICLE TOO MUCH IN ORDER TO AVOID DANGEROUS LEAKAGE OF THE BATTERY ELECTROLYTE.

CAUTION

ELECTROLYTE CONTAINS SULPHURIC ACID: AVOID CONTACT WITH EYES, SKIN AND CLOTHES. IN CASE OF ACCIDENTAL CONTACT, RINSE WITH ABUNDANT WATER AND CONSULT A DOCTOR.

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WARNING

USED BATTERIES ARE HARMFUL FOR THE ENVIRONMENT. COLLECTION AND DISPOSAL SHOULD BE CARRIED OUT IN COMPLIANCE WITH REGULATIONS IN FORCE.

Long periods of inactivity

Battery performance will be poor if the vehicle is not used for a long time. This is the result of the natural phenomenon of battery discharging, and may be due to residual absorption by vehicle components with constant power consumption. Poor battery performance may also be due to environmental conditions and the cleanliness of the poles. In order to avoid difficult starts and/or irreversible damage to the battery, follow any of these steps:

- At least once a month start the engine and run it slightly above idle speed for 10-15 minutes. This keeps all the engine components, as well as the battery, in good working order.

- Take your vehicle to a garage (as indicated in the «Vehicle not used for extended periods» section) to have the battery removed. Have the battery cleaned, charged fully and stored in a dry, ventilated place. Recharge **at least once every two months**.

N.B.

THE BATTERY MUST BE CHARGED WITH A CURRENT EQUAL TO 1/10 OF THE RATED CAPACITY OF THE BATTERY AND FOR NOT LONGER THAN 10 HOURS. CONTACT AN AUTHORISED SERVICE CENTRE TO CARRY OUT THIS OPERATION SAFELY. WHEN REFITTING THE BATTERY MAKE SURE THE LEADS ARE CORRECTLY CONNECTED TO THE TERMINALS.

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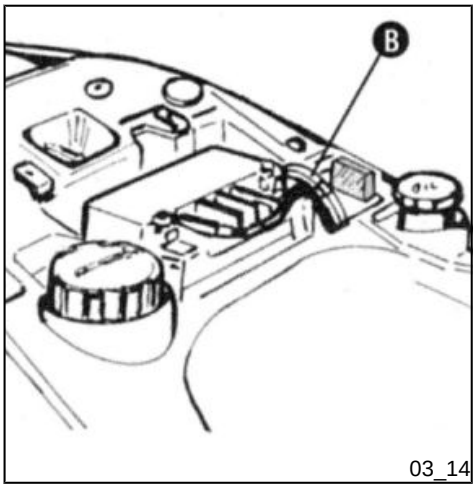
WARNING

DO NOT DISCONNECT THE BATTERY CABLES WITH THE ENGINE RUNNING, THIS CAN CAUSE IRREPARABLE DAMAGE TO THE VEHICLE'S ELECTRONIC CONTROL UNIT.

WARNING

USED BATTERIES ARE HARMFUL FOR THE ENVIRONMENT. COLLECTION AND DISPOSAL SHOULD BE CARRIED OUT IN COMPLIANCE WITH REGULATIONS IN FORCE.



03_14

Fuses (03_14)

The electrical system is protected by a plug fuse «**B**» located to the left of the battery compartment. The ignition system, headlight and the rear light are not fuse-protected. Before replacing a blown fuse, find and solve the problem that caused it to blow. Do not replace the fuse with any alternative form of conductor

CAUTION

IN ORDER TO AVOID DAMAGING THE ELECTRICAL SYSTEM, NEVER DISCONNECT THE WIRING WHILE THE ENGINE IS RUNNING. DO NOT TIP THE VEHICLE TOO MUCH IN ORDER TO AVOID DANGEROUS LEAKAGE OF THE BATTERY ELECTROLYTE.

Characteristic

Fuse (100 cc)

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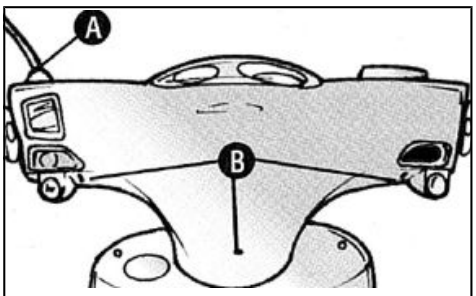
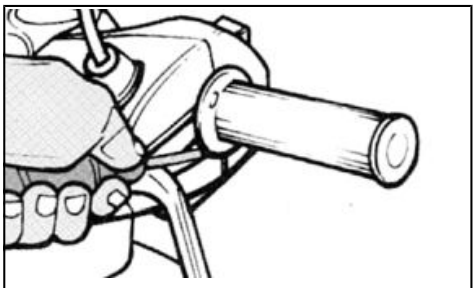


03_15


03_16

Frontlightgroup (03_15, 03_16)

To have access to the headlight bulbs, it is necessary to remove the rear part of the handlebar cover as follows:

Unscrew the rear-view mirror after loosening the screw «**A**» located under the corresponding rubber protection.

Unscrew the 3 screws «**B**» as well.

Position the screwdriver between the front and rear handlebar cover, and levering gently, release the right and left attachments. After removing the bulb holder joints, move the bulb holder counterclockwise and remove it from its seat. To remove the bulb from the bulb holder push and rotate the bulb counterclockwise.

USE A SCREWDRIVER WITH A WORKING END APPROPRIATE TO THE SCREW SIZE TO AVOID DAMAGING THE PAINTWORK.

Electric characteristic

Bulbs

1 bulb per high-/low-beam light

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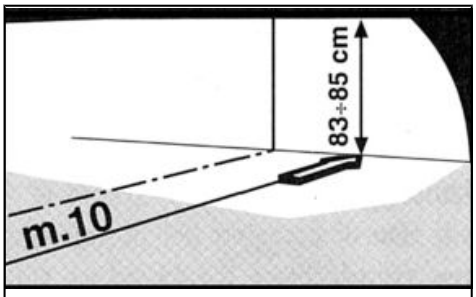


03_17

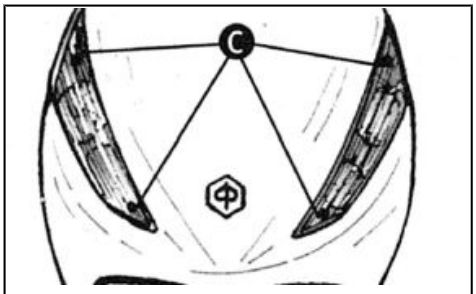
03_18

Headlightadjustment (03_17, 03_18)

Position the unloaded vehicle a flat surface and at 10 m away from a white screen located in the dark. Make sure that the vehicle is perpendicular to the screen. Draw a horizontal line on the screen at a height of 83 ÷ 85 cm from the ground.

Start the engine, turn on the headlight, turn on the low-beam light and orientate it so that the horizontal borderline between the dark and the light area is not higher than the horizontal line drawn on the screen. To remove the headlight, unscrew the screw «**A**» located below it.

Before adjusting the headlight, check that the tyres have the correct pressure as indicated in the «Tyres» section



03_19

Frontdirectionindicators (03_19)

To replace the bulb of the front turn indicators unscrew the 4 fixing screws «**C**». To take out the bulb from the bulb holder, rotate the bulb holder clockwise and remove it from its seat. To remove the bulb, push and rotate it counterclockwise.

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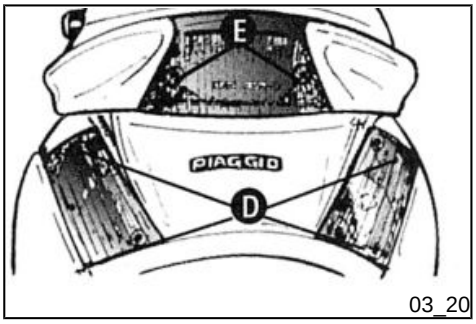
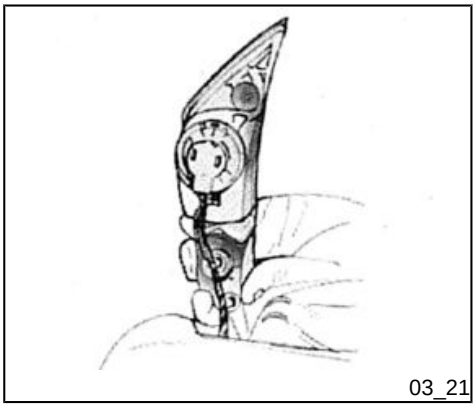


03_20


03_21

Rearopticalunit (03_20, 03_21)

Remove both screws «**E**» and take out the light, then remove the tail light bulbs following the same instructions as for the rear turn indicators; for the stop light, push and rotate the bulb counterclockwise.

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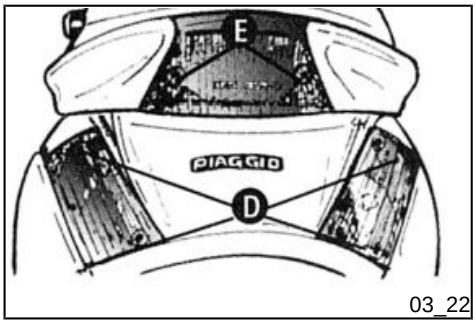
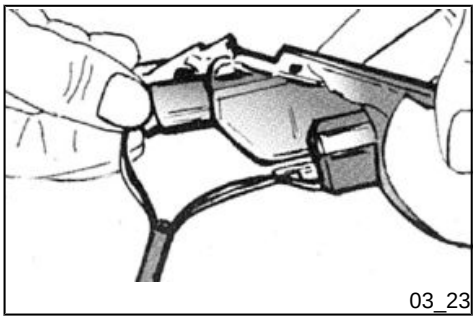


03_22


03_23

Rearturnindicators (03_22, 03_23)

To replace the rear turn indicator lamp, unscrew the 4 fixing screws «**D**» of the glass, to remove it, pull the cable harness and then the bulb.

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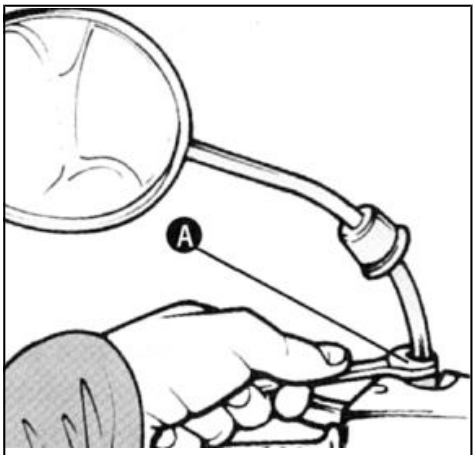
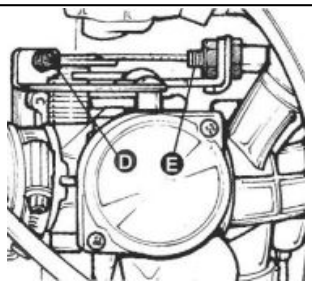


03_24

Rear-viewmirrors (03_24)

To adjust the stem position, lift the rubber protection and loosen the nut «**A**» with an open wrench (opening 13 mm); adjust the stem to the best position and tighten the screw.



03_25

Idleadjustment (03_25)

To adjust the idle speed it is necessary to actuate the screw «**D**» located in the carburettor. It is necessary to actuate on both set screws«**E**» to adjust the clearance of the throttle control transmission. Adjust the idle speed with the rear wheel off the ground (vehicle on stand) and with a warm engine: screw and unscrew the knob-type head screw until reaching a regular rpm: i.e, without letting the engine move the rear wheel.

This operation should be carried out at an Authorised PIAGGIO Service Centre.

SHOULD THERE BE PROBLEMS DURING THE IDLE SPEED ADJUSTMENT OPERATION, IT MAY BE NECESSARY TO ADJUST THE LEVEL OF EXHAUST EMISSIONS (CO). TURN TO AN AUTHORISED PIAGGIO SERVICE CENTRE TO CARRY OUT THESE OPERATIONS.

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CAUTION

WHEN ADJUSTING IDLE SPEED, PAY SPECIAL ATTENTION NOT TO TOUCH THE CATALYTIC MUFFLER TO AVOID BURNS.

Characteristic

Idle speed

1900÷2100 rpm

Engine idle speed (100 cc)

~ 1500 ± 150 rpm

03_26

Frontandreardiscbrake (03_26)

The brake disc and pad wear is automatically compensated, therefore it has no effect on the functioning of the front and rear brakes. For this reason it is not necessary to adjust the brakes. An excessively elastic brake lever stroke may indicate the presence of air in the braking circuit or an irregular brake operation. In this case, particularly considering the importance of the brakes in terms of safety, it is strongly recommended that you take the vehicle to an **Authorised Service Centre** as soon as possible for the appropriate checks.

WARNING

CHECK BRAKE PADS FOR WEAR ON A REGULAR BASIS (AS INDICATED IN THE SCHEDULE MAINTENANCE TABLES). IF THE THICKNESS OF ONE OR BOTH PADS IS IN THE REGION OF 1.5 MM, BOTH PADS MUST BE CHANGED. IT IS RECOMMENDED TO CARRY OUT THIS OPERATION AT AN AUTHORISED SERVICE CENTRE AS SOON AS POSSIBLE.

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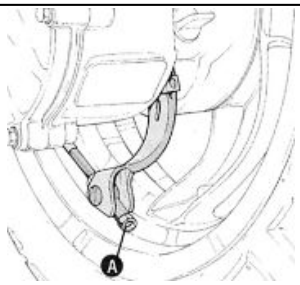
AFTER FITTING NEW BRAKE PADS DO NOT USE THE VEHICLE UNTIL YOU HAVE ACTIVATED THE BRAKE LEVER REPEATEDLY TO POSITION THE PADS AND RESTORE THE LEVER STROKE TO ITS CORRECT POSITION.

CAUTION

BRAKING SHOULD BEGIN AFTER ABOUT 1/3 OF THE BRAKE LEVER STROKE.



03_27

Reardrumbrake (03_27)

Actuate on the set screw «**A**» indicated on the figure bearing in mind that **the wheel must rotate freely** when the control lever is in neutral.

CAUTION

BRAKING SHOULD BEGIN AFTER ABOUT 1/3 OF THE BRAKE LEVER STROKE.

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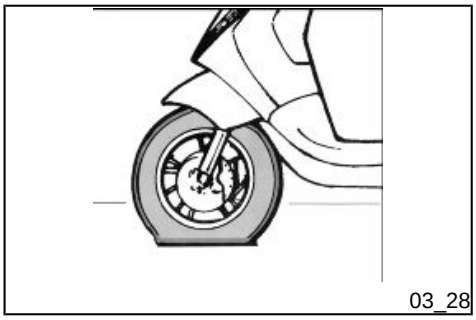

03_28

Puncture (03_28)

The vehicle is equipped with Tubeless tyres (without inner tube). In the event of a puncture, Tubeless tyres - unlike tyres with inner tubes - go flat very slowly, resulting in a greater steering safety. In the event of a puncture, an emergency repair can be carried out using an "inflate and repair" spray can. For a final repair, take your vehicle to an **Authorised Service Centre or Dealer.** The replacement of a tyre involves removing the wheel in question. Take your vehicle to an **Authorised Service Centre or Dealer** for these operations.

CAUTION

TO USE THE "INFLATE AND REPAIR" SPRAY CAN PROPERLY, FOLLOW THE INSTRUCTIONS ON THE PACKAGING.

WARNING

THE WHEELS FITTED WITH TYRES SHOULD ALWAYS BE BALANCED. RIDING THE VEHICLE WITH VERY LOW TYRE PRESSURE OR WITH INCORRECTLY BALANCED TYRES CAN LEAD TO DANGEROUS STEERING VIBRATIONS.

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03_29

delveicolo (03_29)

We recommend carrying out the following operations:
1. General cleaning of the vehicle.

2. With the engine off and the piston at the bottom dead centre position, **remove the spark plug**, and pour 1÷ 2 cm³ of recommended oil through its opening. Press the engine start pedal 3 or 4 times letting the engine perform a few revolutions slowly, then replace the spark plug.

3. Drain off all the vehicle fuel; spread antirust grease on the unpainted metal parts; keep the wheels off the ground, by resting the chassis on two wooden wedges.

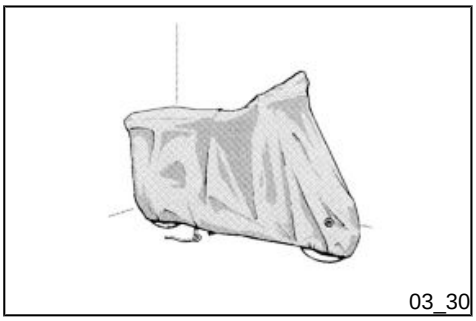
4. For the battery, follow the procedures described in the «Battery» section.

5. Drain the petrol from the carburettor float chamber through the bleed cap.

Recommended products

eni i-Ride PG 5W-40

Synthetic based lubricant for high-performance four-stroke engines.
JASO MA, MA2 - API SL - ACEA A3


03_30

Cleaningthevehicle (03_30)

Use a low pressure jet of water to soften the caked dirt and mud deposited on the painted surfaces. Once softened, sponge off mud and dirt using a car body sponge soaked in a car body shampoo and water solution (2-4% of car shampoo in water). Then rinse with abundant water, and dry with a shammy cloth. For the engine exterior, use petrol, a brush and clean cloths. Petrol can damage paintwork. Remember that any polishing with silicone wax must always be preceded by washing.

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WARNING

To avoid the appearance of oxidations, wash the vehicle every time it is used in certain areas or in special conditions of:

· Environmental / seasonal conditions: use of salt, de-icer chemical products on the road in winter.

· Air pollution: city and/or industrial areas.

· Salinity and humidity of the atmosphere: marine areas, hot and wet weather.

WARNING

. Prevent deposits from remaining on the bodywork, industrial and pollutant residual dust, tar spots, dead insects, bird droppings, etc.

· Do not park the vehicle under the trees. In some seasons, in fact, residues, resins, fruits or leaves may fall from the trees, containing chemicals that are harmful to the paintwork.

CAUTION

DETERGENTS POLLUTE WATER. THEREFORE THE VEHICLE SHOULD BE WASHED IN AN AREA EQUIPPED FOR THE COLLECTION AND PURIFICATION OF THE LIQUIDS USED.

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WARNING

NEVER WASH THE VEHICLE UNDER DIRECT SUNLIGHT, ESPECIALLY IN SUMMER WHEN THE BODYWORK IS STILL HOT, AS THE CAR SHAMPOO MAY DRY BEFORE BEING RINSED OFF, AND COULD DAMAGE THE PAINTWORK. NEVER USE RAGS SOAKED IN PETROL OR DIESEL OIL TO CLEAN THE PAINTED OR PLASTIC SURFACES, IN ORDER TO PREVENT THEM LOSING THEIR SHINE AND MECHANICAL CHARACTERISTICS.

WARNING

WHEN WASHING THE ENGINE WITH A HIGH-PRESSURE WATER JET:

- **ONLY USE FAN SPRAY JETS.**
- **DO NOT PLACE THE NOZZLE CLOSER THAN 60 CM.**
- **DO NOT USE WATER AT TEMPERATURES OVER 40° C.**
- **DO NOT DIRECT THE JETS DIRECTLY TO CARBURETTOR, WIRINGS, SLOT DIFFUSERS ON THE TRANSMISSION COVER AND SCROLL COVER.**

START-UP PROBLEMS

No fuel in tank	Refuel
Filters, jets or carburettor body dirty or clogged.	Contact an **Authorised Service Centre.**
Insufficient battery charge	Kick-start. Recharge the battery.

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IGNITION PROBLEMS

No spark from spark plug. Due to the presence of high voltage, this check should only be carried out by an expert.	Check that the electrodes are properly adjusted (0.7÷ 0.8 mm). Check that the electrodes are clean (use pure petrol and a metal brush or an emery cloth). Check the spark plug insulator: Replace the spark plug if the insulator is cracked or broken. If the spark plug is in good conditions, contact an **Authorised Service Centre.**

LACK OF COMPRESSION

Spark plug loose. Loose cylinder head, worn piston retaining rings. Incorrect valve clearance	Contact an **Authorised Piaggio Service Centre**

HIGH CONSUMPTION AND LOW PERFORMANCE

Air filter blocked or dirty.	Clean with water and shampoo and impregnate with petrol and specific oil («Air filter removal» section)

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INSUFFICIENT BRAKING

Oil on drum or disc. Worn Pads/ Shoes	Contact an **Authorised Service Centre**
Rear brake incorrectly adjusted. Air in the front brakes circuit	Turn to an **Authorised Piaggio Service Centre**

INEFFICIENT SUSPENSION

Oil leak; worn limit switch bumpers; worn shock absorber attachment points	Contact an **Authorised Service Centre**

AUTOMATIC TRANSMISSION PROBLEMS

Deteriorated roller housing or belt.	Contact an **Authorised Service Centre.**

INCREASED EXHAUST NOISE

Deterioration of the SAS system and/or of the tab	Contact an **Authorised Service Centre.**

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PIAGGIO

Chap. 04
Technicaldata

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TECHNICAL DATA

Engine	Single-cylinder, four-stroke
Bore x stroke	39 X 41.8 mm
Engine capacity	49.93 cm^3
Compression ratio	11.5 :1
Ignition advance (before TDC)	8° at 1800 to 2000 rpm 24° at 5000 to 6000 rpm
Spark plug	NGK CR8EB
Alternative spark plug	DENSO U24ESR-NB
Depression carburettor	KEIHIN CVK Ø 18 mm
valve clearance	intake: 0.10 mm - exhaust: 0.10 mm
Max. speed	According to current legislation

DATA

Overall length	1690 mm
Overall width	680 mm
Maximum height	1070 mm
Wheelbase	1250 mm

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Fuel supply	Gravity feed, unleaded petrol, through carburettor.
Lubrication	Engine lubrication with lobe pump (inside the crankcase) controlled by chain and double filter: mesh and centrifugal.
Cooling	Forced-circulation air cooling.
Transmission	With automatic expandable pulley variator, torque server, V-belt, automatic clutch, gear reduction unit.
Fuel tank	In plastic, with a capacity of ~ 7.5 l (approximate value, including ~ 1.2 l reserve)
Front suspension	with telescopic fork.
Rear suspension	fitted with spring and coaxial hydraulic shock absorber. Chassis to engine support with swinging arm.
Front wheel	With 2.15 X 10" rims
Rear wheel	With 3 X 10" rims
Front tyre	100/80-10"
Rear tyre	120/70-10"
Exhaust muffler	Absorption type.
Electronic ignition	Electronic ignition device with incorporated HV coil.
Chassis	Single spar and stamped sheets.

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Front brake	disc brake with hydraulic control (lever on the right side of the handlebar).
Rear brake	Drum brake with expanding shoes, controlled mechanically (lever on the left side of the handlebar).
Kerb weight in running order	84 kg
Maximum load	Driver only.
Engine oil	Capacity: approx. 850 cm³
Engine oil (change)	850 cm³
Rear hub oil	Quantity: approx. ~80 cm³

TECHNICAL DATA 100

ENGINE	Single-cylinder, four-stroke
Bore	50 mm
Compression ratio	10.5 ÷ 11.5 : 1
Ignition advance (before TDC)	8° at 1800 to 2000 rpm 24° at 5000 to 6000 rpm
Spark plug	NGK CR8EB
Depression carburettor	KEIHIN CVK Ø 18 mm
Engine idle speed (100 cc)	~ 1500 ± 150 rpm
valve clearance	intake: 0.10 mm - exhaust: 0.10 mm

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Max. speed	According to current legislation

TECHNICAL DATA 100

Width	735 mm
Length	1880 mm
Maximum height	1150 mm
Wheelbase	1340 mm
Fuel system	Gravity feed, with unleaded petrol (with a minimum octane rating of 95) with carburettor.
Lubrication	engine lubrication with lobe pump (inside the crankcase) controlled by chain and double filter: mesh and centrifugal.
Cooling	Forced-circulation air cooling.
TRANSMISSION	With continuously variable transmission, torque server, V belt, automatic clutch, gear reduction unit.
Main and supplementary fuel tank (including ~ 1.2 l reserve) (100 cc)	~ 7 l
Front suspension	Mechanical telescopic fork with 30 mm stems.
Front tyre size	100/80-10"
Rear tyre size	120/70-10"

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Tyre pressure (100 cc)	Front wheel: 1.3 bar Rear wheel: 1.6 ÷ 1.8 bar
Ignition type	Capacitive discharge electronic ignition, with incorporated HV coil
Chassis type	Welded tubular steel chassis with stamped sheet reinforcements
Front brake	Disc (220 mm diameter) with hydraulic control (lever on the right side of the handlebar).
Rear brake	Drum brake (140 mm diameter) with expanding shoes, mechanically operated
Engine oil	~ 850 cc (recommended oil: SELENIA HI SCOOTER 4 Tech)
Rear hub (100 cc)	approx. 80 cm^3

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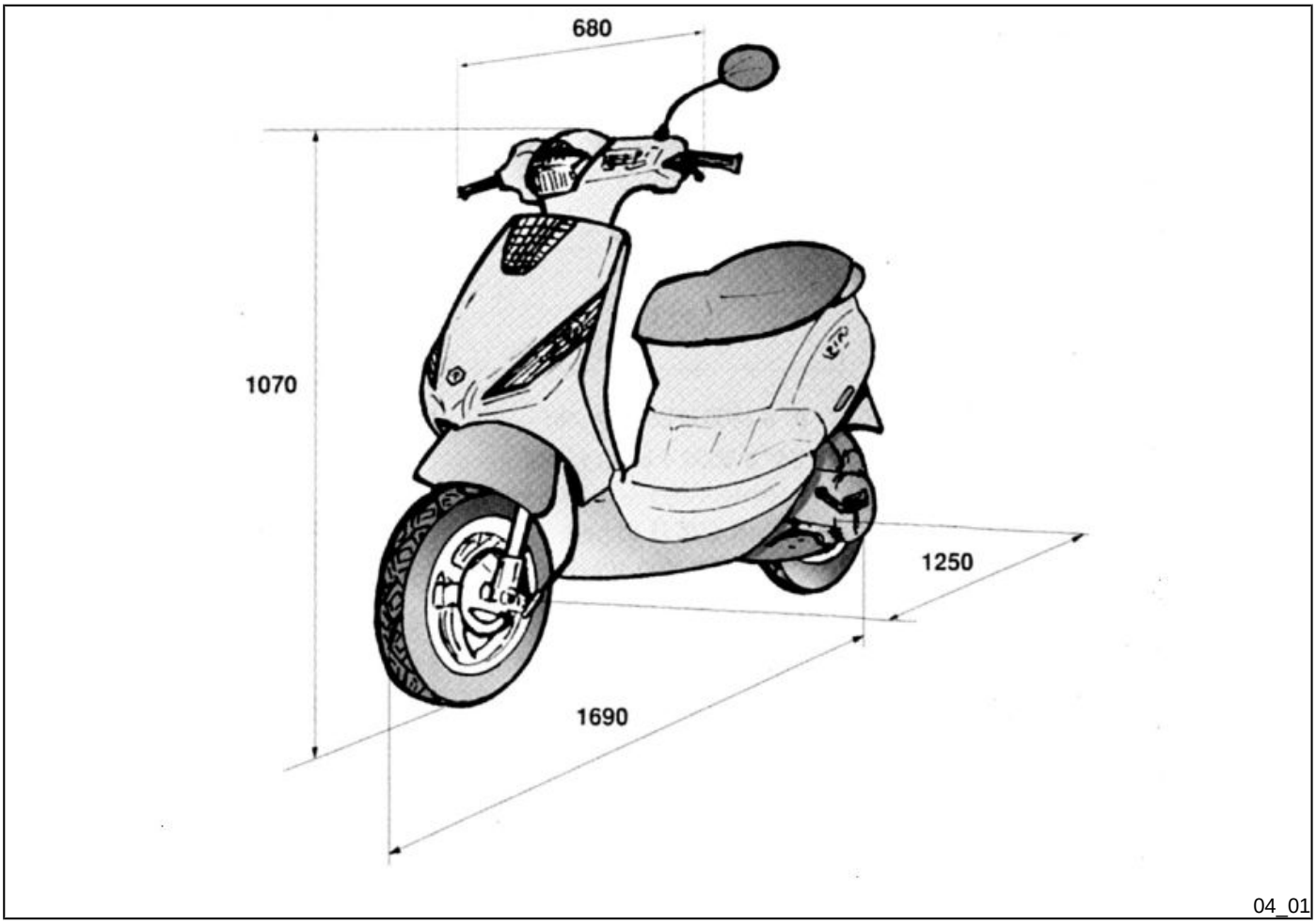

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Toolkit

Wrenches: one box-spanner; one twin screwdriver. The tools are located under the saddle. In order to take out the tools from the case release the lever and remove the clamp.

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PIAGGIO

Chap. 05
Sparepartsandaccessories

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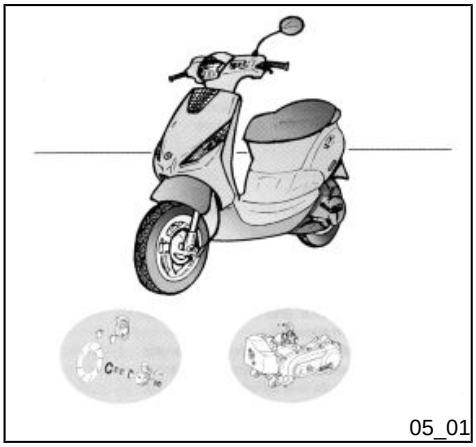

05_01

Warnings (05_01, 05_02)

WARNING

IT IS RECOMMENDED THAT "ORIGINAL PIAGGIO SPARE PARTS" BE USED, AS THESE ARE THE ONLY ONES OFFERING YOU THE SAME QUALITY ASSURANCE AS THOSE INITIALLY FITTED ON THE VEHICLE.

IT SHOULD BE REMEMBERED THAT USING NON-ORIGINAL SPARE PARTS CAUSES YOUR WARRANTY RIGHTS TO EXPIRE.

WARNING

PIAGGIO MARKETS ITS OWN LINE OF ACCESSORIES THAT ARE RECOGNISED AND GUARANTEED FOR USE. IT IS THEREFORE ESSENTIAL TO CONTACT AN AUTHORISED DEALER OR SERVICE CENTRE IN ORDER TO CHOOSE AND FIT ACCESSORIES CORRECTLY. THE USE OF NON-ORIGINAL ACCESSORIES MAY AFFECT THE STABILITY AND OPERATION OF YOUR VEHICLE AND REDUCE SAFETY LEVELS WITH POTENTIAL RISKS FOR THE RIDER.

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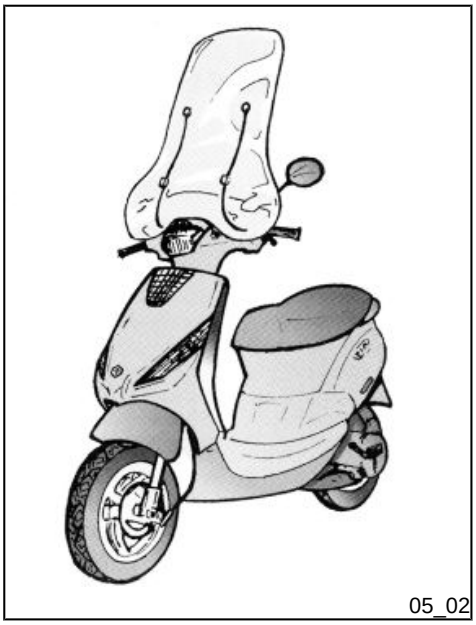

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PIAGGIO

Chap. 06
Scheduledmaintenance

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06_01

Scheduledservicingtable (06_01)

Adequate maintenance is fundamental to ensuring long-lasting, optimum operation and performance of your vehicle.

To this end, a series of checks and maintenance operations (at the owner's expense) have been suggested, which are included in the summary table on the following page. Any minor faults should be reported without delay to an **Authorised Service Centre or Dealer** without waiting until the next scheduled service to solve it.

It is indispensable to have your vehicle serviced to the prescribed intervals of time, even if you have not reached the predicted mileage. Punctual vehicle servicing is necessary for the correct use of the guarantee. For any further information concerning Warranty procedures and 'Scheduled Maintenance', please refer to the 'Warranty Booklet'.

SCHEDULED MAINTENANCE TABLE

Km x 1000	1	3	6	9	12	15	18	21	24	27	30	33	36	39	42	45	48	51	54	57	60
Safety fasteners	I				I				I				I				I				I
Spark plug			R		R		R		R		R		R		R		R		R		R
Drive belt			I		R		I		R		I		R		I		R		I		R
Throttle control	A				A				A				A				A				A
Tyre condition and wear					I				I				I				I				I
Air filter					C				C				C				C				C
Oil filter (mesh)			C		C		C		C		C		C		C		C		C		C
Solenoid filter			C		C		C		C		C		C		C		C		C		C
Valve clearance	I				I				I				I				I				I

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Km x 1000	1	3	6	9	12	15	18	21	24	27	30	33	36	39	42	45	48	51	54	57	60
Electrical system and battery	I		I		I		I		I		I		I		I		I		I		I
Cylinder ventilation system									I								I				
Brake levers	L				L				L				L				L				L
Brake oil level (*)	I		I		I		I		I		I		I		I		I		I		I
Hub oil level	R		I		R		I		R		I		R		I		R		I		R
Engine oil	R	I	R	I	R	I	R	I	R	I	R	I	R	I	R	I	R	I	R	I	R
Brake Pads/Shoes	I		I		I		I		I		I		I		I		I		I		I
Tyre pressure	I		I		I		I		I		I		I		I		I		I		I
Headlight					A				A				A				A				A
Vehicle road test	I		I		I		I		I		I		I		I		I		I		I
Idle speed	A				A				A				A				A				A
Odometer gear					L				L				L				L				L
CVT rollers			I		R		I		R		I		R		I		R		I		R
Suspension					I				I				I				I				I
Steering	A				A				A				A				A				A
Transmission					L				L				L				L				L

***I**: CHECK AND CLEAN, ADJUST, LUBRICATE OR REPLACE IF NECESSARY.*

***C**: CLEAN, **R**:REPLACE, **A**: ADJUST, **L**:LUBRICATE*

** Replace every 2 years*

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Scheduledservicingtable (06_01)

Adequate maintenance is fundamental to ensuring long-lasting, optimum operation and performance of your vehicle.

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It is indispensable to have your vehicle serviced to the prescribed intervals of time, even if you have not reached the predicted mileage. Punctual vehicle servicing is necessary for the correct use of the guarantee. For any further information concerning Warranty procedures and 'Scheduled Maintenance', please refer to the 'Warranty Booklet'.

EVERY 3000 KM
Engine oil - level check/ top-up
Air filter - clean

EVERY 2 YEARS
Brake fluid - change

AFTER 1,000 KM OR 4 MONTHS
Hub oil - change

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Valve clearance - check
idle rpm (*) - adjustment
Throttle lever - adjustment
Steering - adjustment
Brake levers - greasing
Brake pads - check for condition and wear
Brake fluid level - check
Safety fasteners - check
Electrical system and battery - check
Tyre pressure - check
Vehicle test and brake test - Road test

(*) See instructions in «Idle speed adjustment» section

At 6000 km or 12 months, 18000, 30000, 42000, 54000 and 66000 km

Engine oil - change
Hub oil level - control
oil filter - cleaning
CVT rollers - check or replacement
Brake pads - check for condition and wear
Brake fluid level - check

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Electrical system and battery - check
Tyre pressure and wear - check
Tyre pressure - check
Vehicle test and brake test - Road test

At 12000 km or 24 months and 60000 km

Engine oil - change
Hub oil level - control
Spark plug / electrode gap - replacement
oil filter - cleaning
idle rpm (*) - adjustment
Throttle lever - adjustment
CVT rollers - check or replacement
Drive belt - replacement
Odometer gear - greasing
Steering - adjustment
Brake levers - greasing
Brake pads - check for condition and wear
Brake fluid level - check
Transmission - lubrication
Safety fasteners - check

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Suspension - check
Electrical system and battery - check
Headlight - adjustment
Tyre pressure and wear - check
Tyre pressure - check
SAS box (sponge) (**) - cleaning
Vehicle test and brake test - Road test

(*) See instructions in «Idle speed adjustment» section

(**) See regulations in the «Secondary air system» section

After 24000 km and 48000 km

Engine oil - change
Hub oil - change
Spark plug / electrode gap - replacement
oil filter - cleaning
Valve clearance - check
idle rpm (*) - adjustment
Throttle lever - adjustment
CVT rollers - check or replacement
Drive belt - replacement
Cylinder ventilation system - check

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Odometer gear - greasing
Steering - adjustment
Brake levers - greasing
Brake pads - check for condition and wear
Brake fluid level - check
Transmission - lubrication
Safety fasteners - check
Suspension - check
Electrical system and battery - check
Headlight - adjustment
Tyre pressure and wear - check
Tyre pressure - check
SAS box (sponge) (**) - cleaning
Vehicle test and brake test - Road test

(*) See instructions in «Idle speed adjustment» section

(**) See regulations in the «Secondary air system» section

AFTER 36000 KM

Engine oil - change
Hub oil level - control
Spark plug / electrode gap - replacement

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oil filter - cleaning
idle rpm (*) - adjustment
Throttle lever - adjustment
CVT rollers - check or replacement
Drive belt - replacement
Odometer gear - greasing
Steering - adjustment
Brake levers - greasing
Brake pads - check for condition and wear
Brake fluid hoses - replacement
Brake fluid level - check
Transmission - lubrication
Safety fasteners - check
Suspension - check
Electrical system and battery - check
Headlight - adjustment
Tyre pressure and wear - check
Tyre pressure - check
SAS box (sponge) (**) - cleaning
Vehicle test and brake test - Road test

(*) See instructions in «Idle speed adjustment» section

(**) See regulations in the «Secondary air system» section

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After 72000 km

Engine oil - change
Hub oil - change
Spark plug / electrode gap - replacement
oil filter - cleaning
Valve clearance - check
idle rpm (*) - adjustment
Throttle lever - adjustment
CVT rollers - check or replacement
Drive belt - replacement
Cylinder ventilation system - check
Odometer gear - greasing
Steering - adjustment
Brake levers - greasing
Brake pads - check for condition and wear
Brake fluid hoses - replacement
Brake fluid level - check
Transmission - lubrication
Safety fasteners - check
Suspension - check
Electrical system and battery - check
Headlight - adjustment

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Tyre pressure and wear - check
Tyre pressure - check
SAS box (sponge) (**) - cleaning
Vehicle test and brake test - Road test

(*) See instructions in «Idle speed adjustment» section

(**) See regulations in the «Secondary air system» section

RECOMMENDED PRODUCTS TABLE

Product	Description	Specifications
AGIP GEAR SAE 80W-90	Lubricant for gearboxes and transmissions.	API GL-4
AGIP BRAKE 4	Brake fluid.	Synthetic fluid SAE J 1703 -FMVSS 116 - DOT 3/4 - ISO 4925 - CUNA NC 956 DOT 4
eni i-Ride PG 5W-40	Synthetic based lubricant for high-performance four-stroke engines.	JASO MA, MA2 - API SL - ACEA A3
AGIP FILTER OIL	Special product for the treatment of foam filters.	-
AGIP GREASE MU3	Yellow-brown, lithium-base, medium-fibre multipurpose grease.	ISO L-X-BCHA 3 - DIN 51 825 K3K -20
AGIP GP 330	Water repellent stringy calcium spray grease.	R.I.D./A.D.R. 2 10°b) 2 R.I.Na. 2.42 - I.A.T.A. 2 - I.M.D.G. class 2 UN 1950 Page 9022 EM 25-89

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TABLE OF CONTENTS

B
Battery: *34*

C
Checks: *16*

F
Fuses: *36*

I
Identification: *14*

K
Keys: *13*

M
Maintenance: *25*

P
Puncture: *44*

R
Refuelling: *16*

T
Toolkit: *58*
Tyres: *29*

V
Vehicle: *7*

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PIAGGIO

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